

Design of a Master Console for a MR-Conditional Teleoperated Right Heart Catheterisation Robot

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ABSTRACT

Teleoperated robotic catheterisation systems have been developed to address the limitations of manual right heart catheterisation under magnetic resonance imaging, allowing operators to perform operations remotely where in-bore manipulation is challenging. However, existing systems predominately employ joystick-based control interfaces, requiring clinicians to decompose catheter manipulation movements into discrete control inputs. This limits clinicians' ability to leverage their prior experience, as the control mapping diverges from established practice, resulting in an additional learning curve and increased cognitive load during operation. This paper proposes a master console design for a current MR-conditional robotic system for right heart catheterisation. To create an intuitive master console, a catheter-replica controller approach is adopted. The master console consists of a mechanical design with a cylindrical handle coupled to a shaft with a machined drive flat, combined with an auto re-centring belt drive mechanism. The configuration provides two degrees of freedom input for infinite, continuous translation and rotation through a single unified interface that satisfies the motion requirements for the robotic operation of a Swan-Ganz catheter. The contribution of this paper lies in the development of a master console that preserves the natural push-pull and axial rotation motions of direct manual catheter handling. Three experiments validates the infinite translation with the re-centring mechanism, multi-turn rotation accumulation, and commanded target position navigation capability across the robotic system's operational envelope.

Keywords: Robotic catheterisation, master console, MR-conditional, human-machine interface

1 Introduction

Teleoperated robotic systems have become widely adopted across a range of minimally invasive surgeries and procedures, enhancing precision, efficiency and safety. These advancements have led to improved clinical outcomes, with patients on average experiencing reduced pain, decreased blood loss and shorter hospital stays [1]. Despite this progress, there is room for further refinement of the physical human-machine interface through which clinicians interact with these systems. Many current surgical and medical robotic systems rely on joystick-based control consoles. While this method of control is effective and well-established, these interfaces may diverge from the experience and procedural habits clinicians develop through traditional manual practice [2]. This highlights an opportunity to design more intuitive control interfaces, especially master manipulators that better reflect established techniques while incorporating human factors considera-

tions [3]. Such approaches may enable more natural and precise translation of the clinician's intent, further optimising overall procedural performance.

This work builds on an existing magnetic resonance (MR)-conditional, two-degree-of-freedom (DoF), teleoperated robotic system for right heart catheterisation (RHC) [4] developed by Wang et al. at University College London (UCL). The system supports teleoperation of a Swan-Ganz catheter while the patient remains inside the MRI scanner bore. It effectively addresses the key challenges of manual catheterization within a confined MRI bore, where limited space restricts access to catheter entry points and prolonged operation can lead to fatigue and back strain. The aim of this study is to develop an intuitive master console for the UCL catheterisation robotic system, with a focus on usability and the motion input requirements for controlling a Swan-Ganz catheter. To achieve this, a catheter-replica cylindrical handle ap-

proach is adopted, with the proposed interface design capable of capturing the two-DoF translational and rotational inputs in a manner consistent with established manual catheter manipulation.

2 Literature review

2.1 Clinical Workflow and Motion Requirements for Robotic Right Heart Catheterisation

RHC is a very common and standard diagnostic procedure for measurement of key data such as hemodynamic pressures, oxygen levels, and blood flow within the right side of the heart. The procedure involves the insertion of a catheter through a vein usually near the neck, chest or groin area [5]. The catheter is then navigated through the inferior vena cava (IVC), a large vein that returns deoxygenated blood to the heart, then through the heart chambers from right atrium (RA) to the right ventricle (RV), and finally to the pulmonary artery (PA) where the data will be measured.

For this procedure, a Swan-Ganz pulmonary artery catheter is employed. It is a thin, flexible tube with a pre-curved geometry and an inflatable balloon at its tip. The inflation of the balloon allows the catheter to be carried by the blood flow through the veins. In combination with its curved geometry, the catheter can be rotated axially to orient the curved tip, navigating through the anatomical route while pulled passively by the blood flow into the pulmonary artery. This method of flow-directed navigation reduces the degree of freedom required for the steering of the catheter to two, which is the axial translation and axial rotation.

From empirical characterisation of manual Swan-Ganz RHC process performed by clinicians on patient-specific cardiac phantoms, the unidirectional rotational inputs ranges from 525° to 1105° and translational displacements between 37 and 96 mm for each step [4]. These motions were not applied independently, but simultaneous during manipulation. This establishes the minimum motion envelope that the robotic system and master console must support for the clinical procedure. The large rotational input range necessitates effectively infinite continuous rotation, while the translational input must provide a stroke length that exceeds the maximum translational dis-

placement per step to allow uninterrupted manipulation. The system must also support simultaneous rotational and translational input and motion. Additionally, the system must provide control for the inflation and deflation of the balloon at the catheter tip for flow-directed navigation. Lastly, the system must ensure compatibility with the MRI environment for the procedure to be performed safely under MRI guidance.

2.2 Current State of Robotic Catheterisation Master Manipulator

Current teleoperated catheter robotic systems are employed in a wide range of clinical applications within interventional cardiology, electrophysiology and vascular neurology. While this paper focuses on RHC, other robot catheter systems with different clinical objectives share a similar challenge for the design of the master control interface. They provide insight into the current state of user interface and control strategies.

2.2.1 UCL MR-conditional RHC Console

The existing master console of the UCL RHC robot that this paper builds upon serves as the design baseline. The console is joystick-based, with four separate joysticks governing the control of the catheter movements. Two joysticks are used for the pure axial rotation and translation, while a third joystick enables simultaneous rotation and translation, and the fourth joystick controls catheter balloon inflation and deflation. The separate joystick inputs rather than a unified interface that controls both DoF simultaneously represents the principal usability limitation of the design. This arrangement forces the clinician to decompose manipulation gestures into discrete control channels, diverging from the natural hand motions of manual catheter handling and thus increasing the cognitive load during control.

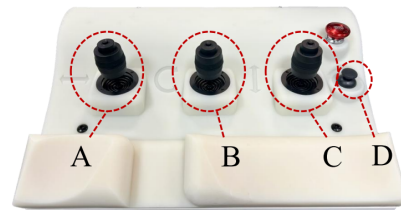


Figure 1: The four joystick console configuration of the UCL RHC robot, with each of the joysticks governing translational, rotational, simultaneous movements and balloon inflation [4]

2.2.2 Joystick-Based Systems

Many commercially available catheter robotic systems employ a joystick-based control, notably the Sensei X and Magellan robotic system from Hansen Medical [6][7] and the CorPath GRX system from Corindus Vascular Robotics [8]. The Sensei X system targets cardiac electrophysiology ablation, with catheter navigation performed via a specialized 3D joystick with motion control technology, which translates the clinician's hand motions to catheter movement commands. The Magellan robotic system is considered an improvement from the Sensei X system, and is used for endovascular catheterisation procedures. It employs a 7-DoF joystick with an open console configuration that provides better postural freedom than Sensei X. The CorPath GRX system is used for percutaneous coronary intervention, with its console featuring three separate joysticks to control the guide catheter, guidewire and balloon. In addition, a touchscreen interface enables millimetre-scale precision in the manipulation of the three components. While these commercial systems are well established and used routinely on their targeted clinical procedures, their inputs remain an abstracted representation of the catheter manipulation process. Thus, there are alternative control methods explored in both commercial and research systems with the goal of improving the control mapping.

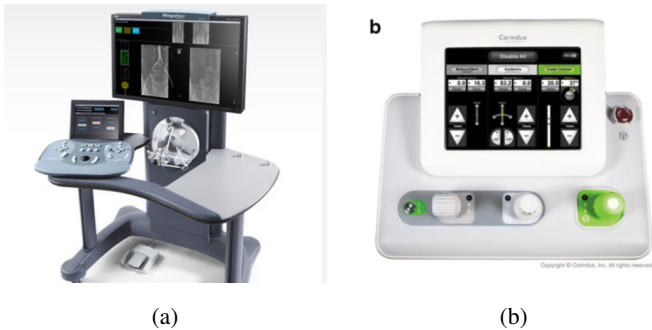


Figure 2: Commercial joystick-based robotic catheter consoles. (a) The Hansen Medical Sensei X operator workstation, with a 3D joystick in the centre alongside a dedicated control panel to its left [6]. (b) The Corindus CorPath GRX console, composing of three independent joysticks for control of the guide catheter, guidewire, and balloon, integrated with a control panel for precise controls [8].

2.2.3 Alternate Systems

The Niobe ES is a remote magnetic navigation system [9] that manipulates a magnetically steerable catheter using two magnet pods on both sides of the medical-

imaging table, controlled through a computer interface. Although this system suggests a new steering method completely different from manual catheter manipulation, a study shows that the system is intuitive and easy to use, where novice operators with no prior catheter ablation experience achieves proficiency in a rapid learning curve through training with the Niobe ES [10]. This suggests alternative steering mechanisms that differs significantly from manual catheter manipulation can still provide effective and user-friendly control. However, this method can't be replicated in a MR environment due to the use of ferromagnetic materials, and is only compatible with fluoroscopy imaging.

Conversely, there are also systems that were developed explicitly to replicate the experience of manual catheter manipulation, including the Amigo Remote Catheter System from Catheter Precision [11] and Cathbot at Imperial College London [12]. The master console of the Amigo system is designed as a replica of a catheter handle, where axial translation and rotation is controlled by sliding and rotating the handle similar to a real catheter. A thumb wheel control on the handle is also used for handling tip steering. Similarly for Cathbot, a cylindrical handle is used to preserve the natural push pull and rotation motion. This approach allows for better mapping of control inputs by preserving the kinematic characteristics of manual catheter manipulation. This has the potential to provide a more accurate translation of the operator's intent with minimal learning curve. These systems have inspired the design of the master console proposed in this paper.

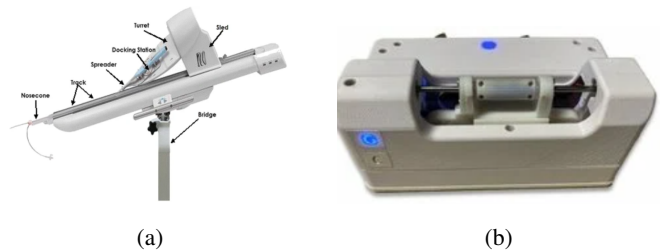


Figure 3: Master consoles with catheter replica approaches. (a) The Catheter Precision Amigo Remote Catheter System Console, featuring a handheld controller shaped to resemble a catheter handle [11]. (b) The Imperial College London Cathbot console, with a cylindrical handle design [12].

At the current stage, there is a progression from the predominately joystick-based landscape towards a heavier focus on specialized master consoles. As these medical robotic systems can't be fully automated due to the

variability between patients, Procedural success continues to rely heavily on surgeon skill and experience. Consequently, improving the design of the master console presents a great approach to improving procedural outcomes of these robotic systems.

3 System Design and Implementation

The master console design in this paper is inspired by the catheter handle replica approach. The following sections detail the functional requirements, and the usability and human factors considerations associated with this design.

3.1 Design Requirements

3.1.1 Functional and Motion Requirements

The design of the master console for the UCL catheterization robot must take into consideration the existing robot's motion kinematics, control mapping and the clinical demands of right heart catheterisation. Therefore, the master console should meet the following functional requirements:

1. The console must support the two degrees of freedom for axial translation and axial rotation necessary for Swan-Ganz catheter manipulation.
2. The console must support continuous, infinite and simultaneous rotation and translation input.
3. Axial translation must span a minimum stroke length of 100mm in each direction from home position for each push and pull step without requiring operator repositioning.
4. Minimum Resolution of the translational sensing must exceed the resolution of the slave robot's actuator, which is a active-passive roller system with a 1 mm catheter groove.
5. The console must include a dedicated input to command balloon inflation and deflation operable while controlling the catheter movement.
6. The console must be able to communicate from the control room to the slave robot within the MRI room via long, MR-conditional cables

3.1.2 Usability and Human Factors Considerations

Effective catheter teleoperation requires the master console to include design choices that minimise the cognitive and physical burden on the operator. Hence, aside from the functional requirements, the following usability and human factors must also be considered to design an ergonomic control console:

1. The control mapping shall replicate the the natural gestures of direct bedside catheter manipulation closely through mechanical design, such that experienced clinicians can operate the console with minimal retraining.
2. The catheter replica or cylindrical handle dimension shall fall within a range established for a comfortable precision grip with gloved hands.
3. The spatial layout for the handle grip and balloon inflation and deflation button shall be positioned within close reach without requiring change in grip posture.
4. The handle orientation and console height should support a seated operating posture, ensuring sustained comfort over the duration of a RHC procedure.

3.2 Design Concept and Rationale

The design in this paper departs from the prevalent paradigm of using a joystick as the primary control input for robotic catheter teleoperation, exploring alternative approaches that better leverage the experience and manipulation techniques that clinicians develop through direct bedside catheterisation. The present design takes the approach of a single integrated grip that encodes both degrees of freedom with a cylindrical handle controller. The cylindrical handle is mounted on a shaft such that it can be freely pushed and pulled along the shaft axis with a stroke length of 100 mm for translation input, while rotationally coupled to the shaft for measurement of rotation. This configuration allows the operator to use the same natural catheter manipulation hand motions for advance, retract and rotate commands to control the catheter.

The requirement for rotation is achieved through the cylindrical handle that allows infinite and continuous rotation. As for the infinite translation requirement, an auto re-centring mechanism is implemented to return the cylindrical handle to a home position after each step. This is achieved through a control algorithm where the sensors

keep track of touch sensing on the handle and catheter axial displacement along the shaft. Upon clinician releasing their grip on the handle, a belt-drive mechanism connected to the cylindrical handle brings the carriage back to its neutral position on the shaft.

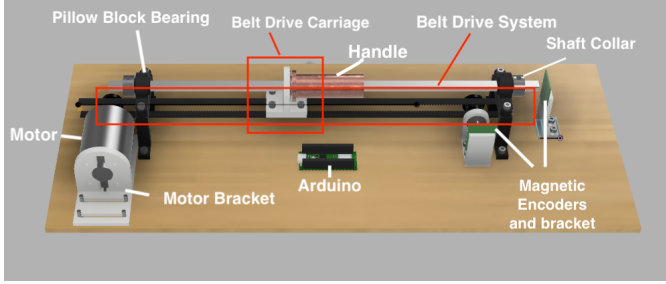


Figure 4: CAD model of the proposed master console design, featuring a configuration with cylindrical handle mounted on shaft, supported at both ends by riser blocks and pillow block bearings, constrained by shaft collars, and with handle connected to a belt-drive carriage that is actuated by the belt drive system labelled by the red boxes.

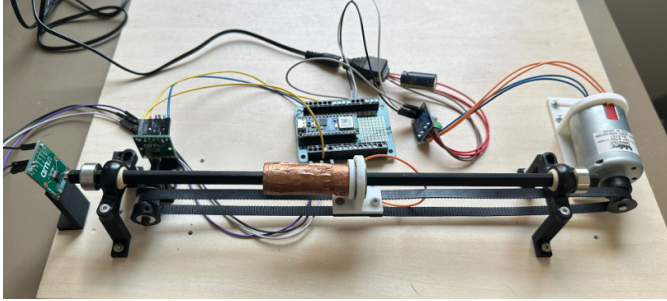


Figure 5: Manufactured prototype of the master console from the CAD design, displaying the overall assembly of the cylindrical handle, shaft, belt-drive system, and electronics.

3.3 Mechanical Design of the Master Console

The master console is built around a primary linear axis defined by a 8mm diameter aluminium round shaft with a machined drive flat, 350 mm in length, supported at each end by pillow block bearings, lifted by riser blocks and constrained by shaft collars. This configuration helps fix the shaft at a suitable height for comfortable operation while constraining it to pure rotation. The cylindrical handle is a 3D-printed poly lactic acid (PLA) part with a corresponding D-profiled bore, dimensioned to fit the machined flat of the shaft. The clinician can slide the handle freely along the shaft to perform push and pull motions, while the cylindrical shaped handle allows continuous rotation, which is transmitted directly to the shaft through the D-bore coupling. The handle is centred midway along the shaft at a home position, providing a stroke length of 100 mm in either direction. Together, this provides the

two degrees of freedom input with a suitable operation range through a single unified input interface.

3.4 Rotational and Translational Sensing

The rotational position of the cylindrical handle is measured by a 14-bit rotary position encoder (ams OSRAM AS5048A) via contactless magnetic sensing. The encoder is mounted approximately 1 mm from one end of the primary shaft, where a small diametrically magnetised permanent magnet is fixed to the shaft end. The encoder has a high resolution of around 0.022° , which is equivalent to 16,384 positions per revolution, providing a very precise measurement with minimal uncertainty. The contactless sensing principle of the magnetic sensor contributes to the infinite rotation input capability of the handle. As the encoder provides absolute position over a single revolution only, the firmware implements a running-total approach that accumulates the net angular displacement, preserving the full resolution of the sensor across multi-turn rotation movements.

The translational position of the handle is measured with a second AS5048A encoder, mounted on the idler pulley shaft of the belt-drive mechanism. As the carriage is mechanically coupled to the cylindrical handle for the auto re-centring mechanism, they share the same linear displacement. So, the encoder measures the angular position of the pulley shaft, which is converted into linear displacement by the firmware using the pulley geometry. For the 20-tooth, 2 mm pitch timing belt pulley, one full revolution of the pulley shaft represents 40 mm in linear motion. This provides a theoretical linear resolution of 2.44 μm for the 14-bit resolution of the encoder. However, in practice, the positional accuracy is dominated by mechanical factors in the belt-drive system such as belt elasticity and transmission backlash. The accumulated translational displacement will be displayed to the operator to indicate insertion depth.

3.5 Auto re-centring mechanism

To implement the infinite translation requirement, a belt drive system is used in combination with grip sensing on the cylinder handle and the translational position encoder. When grip is detected during clinician manipulation, the encoder sends the translational command signals to the

slave robot, while the position of the cylinder handle relative to the shaft is sampled continuously. When the grip is released, the carriage connected to the cylindrical handle is brought back to the home position by the belt drive system. The belt drive carriage and the cylindrical handle are axially coupled through a ball bearing interface, such that the translational motion is shared while rotational independence is preserved. This allows the cylindrical handle to continue to rotate freely, without imparting any rotational load to the carriage under the belt drive constraint.

The belt drive is actuated by a brushed DC motor (NIDEC Advanced Motor, 7.2 W, 12 V, 4700 rpm), driving a closed-loop GT2 timing belt via a 20-tooth pulley mounted on the motor shaft, with another idler pulley that keeps the belt under tension. The GT2 tooth profile minimises belt slip to provide better precision. The translating carriage consists of two 3D-printed components where the top and bottom piece clamps the timing belt in between with a bolted joint. With the belt drive oriented parallel to and directly placed beneath the primary shaft, the carriage is co-axial with the primary shaft axis. This allows the motor shaft to actuate the belt drive and displace the carriage linearly along the shaft. The motor drive current is supplied by a H-bridge motor driver breakout board (Adafruit DRV8871), accepting PWM control signals from the microcontroller (Arduino Nano ESP32) and delivering sufficient drive current to actuate the motor accordingly. An electrolytic capacitor (Panasonic 1000 μ F, 35 V) is placed across the motor power supply rail to suppress back-EMF voltage generated when motor brakes and changes direction. The 12 V power supply is provided by an plug-in AC/DC adapter (RS PRO 24 W, 12 V DC, 2 A), connected via a cable-mount female socket (RS PRO CLB-JL, 5A, 12 V DC).

Grip sensing is implemented through capacitive touch sensing integrated to the outer surface of the cylindrical handle. Conductive copper metallic tape is applied to the grip surface that serves as the sensing electrode. The signal from the copper tape is routed to the capacitive input pin of the Arduino Nano ESP32, which supports capacitive touch sensing. To accommodate continuous handle rotation, a slip ring is integrated to maintain the electrical contact regardless of rotation of the handle. The capacitive measurement detects the presence of the operator's hand in close proximity with the tape surface. To

ensure signal integrity, a short time-based filter is applied in the control algorithm, requiring the capacitive signal to remain continuously above the detection threshold for a short hold duration to confirm the grip state. Only when deliberate clinician grip is confirmed, the translational and rotational inputs are forwarded as command signals to the slave robot, serving as a safety interlock that prevents unintended handle contact from actuating the system.

3.6 Control Interface and Mapping

The console provides the two sensor outputs to the slave robot controller for the axial handle displacement in millimetres and the handle rotation in degrees. The balloon inflation and deflation command is transmitted as a single Boolean signal through the button state. The console operates with a 1:1 input to output motion scaling ratio, where the handle displacement and rotation corresponds to the same distance and angle in the catheter movement. This direct mapping preserves the natural sense of position to replicate the experience of bedside catheter handling.

For the simultaneous motion requirement, the operator can perform both translation and rotation simultaneously with the control handle. With the two separate encoders sampling and transmitting input data independently, the movements in both degrees of freedom can be captured without degradation. The Arduino Nano ESP32 executes the control loop at 100 Hz, giving a maximum command latency of 10 ms, which is well below the human perception of delay threshold window of 100-200 ms for robot teleoperation [13].

3.7 System Architecture and Workflow

The master console is placed in the control room, located outside of the MRI room, which includes the cylindrical handle and shaft assembly, timing belt return mechanism and the balloon inflation button. The Arduino Nano ESP32 microcontroller reads all sensor signals, implements the grip state and auto re-centring control logic for the belt drive motor, and transmits control handle position and rotation values to the slave robot through shielded cables. The slave robot then interprets these commands and actuates the catheter through the catheter robot's drive modules.

The complete signal flow and operator workflow dur-

ing a MR-guided RHC catheterization procedure is listed as follows. First (1), prior to the procedure, venous access should be established and the catheter introduced through the access sheath, such that the teleoperated robotic phase begins at the venous entry point. The cylindrical handle should be manually positioned at the home position midway along the shaft, such that the full 100 mm stroke is available in both directions. Then, (2) the clinician starts the operation by gripping the cylindrical handle, which is confirmed by the touch sensor and capacitive detection threshold. Upon confirmed grip (3), the microcontroller begins sampling both encoders, where the displacement and rotation values are sent to the slave robot as command signals to actuate the catheter. After each translation step (4), the clinician releases their grip, the transmission of control signal to the slave robot stops, allowing the belt-drive motor to return the carriage with the cylindrical handle to the home position. Steps 2 to 4 are repeated iteratively to advance and navigate the catheter through the anatomical path to the right atrium. Upon reaching the right atrium (5), the clinician presses the balloon inflation button to inflate the balloon tip, allowing the catheter to be guided passively by the blood flow while the clinician continues to steer the catheter with the control handle. At suitable positions within the pulmonary artery (6), haemodynamic measurements are taken while the balloon is still inflated. Then, the balloon is deflated before other diagnostic measurements. Lastly (7), with confirmation that the balloon is fully deflated, the clinician withdraws the catheter employing a similar process to steps 2 to 4, but with a reversed handle translation direction until the catheter is fully retracted.

4 Experimental set-up

To validate the functional performance of the master console, a series of characterisation experiments were conducted to evaluate the core requirements on translation, rotation and navigation. These experiments were conducted using a software validation setup, where the inputs from the master console were streamed to a real-time visualisation interface for a simulated position display. This approach enables testing of the master console sensing, actuation, and control logic before it is integrated with the slave robot. Experiment 1 evaluates the consistency of the

translational encoder readings to pre-marked target positions following auto re-centring. Experiment 2 validates the requirement for continuous, multi-turn handle rotation beyond a single revolution. Experiment 3 assesses the system's capability to be navigated to and held at target positions, including targets requiring multiple translation steps.

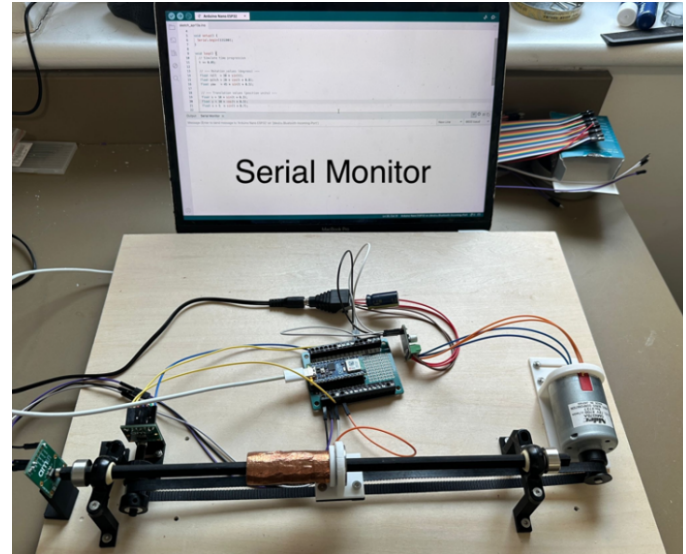


Figure 6: Experimental Setup with prototype connected to a real-time serial monitor display for the current translational and rotational values.

Prior to testing, the system is calibrated by recording the encoder readings by moving it to known linear displacement and rotation positions. Then, the control algorithm can adjust the scaling factor to achieve the 1:1 input to output mapping. This ensures the accuracy of the system, where the movement of the cylindrical control handle translates directly to the same movement for the catheter. Hence, the proposed experiments focus on the measurement repeatability and validation of functional requirements.

4.1 Experiment 1 - Translational Repeatability Characterisation

1. Experiment Objective: In this experiment, the repeatability of the translational reading is evaluated, with particular focus to whether the auto re-centring mechanism introduced systematic positional inconsistency across successive trials.
2. Experiment Protocol: Physical target positions at 10, 50, 100mm from the home position in both directions

of the shaft axis were marked on the shaft using a caliper referenced from the home position. For each target position, the operator displaced the handle to the marked position, where the encoder readings for displacement were recorded via the serial monitor output in the Arduino IDE. The operator's grip was then released, allowing the auto re-centring mechanism to return the carriage and control handle back to the home position. This procedure was repeated for five trials at each target position, for a total of 30 observation across all target positions and trials. The error and standard deviation of the readings was then computed to evaluate the repeatability.

4.2 Experiment 2 - Multi-Turn Rotational Accumulation Verification

1. Experiment Objective: The handle rotation is mechanically coupled to the shaft, and the encoder magnet is fixed to the shaft end. Consequently, the handle rotation is transmitted directly to the encoder without slip and backlash or repeatability concerns in comparison to the translational sensing. Hence, the rotational precision depends on the operator control. The validation focus on this experiment is the control algorithm accumulation logic of rotation across multiple revolutions, as the AS5048A encoder only provides absolute position for a single revolution.
2. Experiment Protocol: A reference mark was applied to the handle. The operator rotated the handle continuously in one direction for five full revolution, with each revolution confirmed by the reference mark returning to the starting position. This procedure was also repeated in the reverse direction. The accumulated angle was recorded and compared against the expected value to verify that the multi-turn revolution angle in both directions are accumulated correctly.

4.3 Experiment 3 - Commanded Target Position Acquisition

1. Experiment Objective: This experiment was designed to evaluate the navigational performance of the master console in reaching arbitrary positions within the system's operational envelope under simultaneous translational and rotational inputs.

2. Experiment Protocol: The translational and rotational encoder outputs from the master console were streamed in real time to a visual interface, providing feedback on the console's current position and rotational angle to the operator. Ten targets were defined across two groups: five targets with translational displacements within a single-stroke range of 100 mm, and five targets in the range of 100 to 200 mm, requiring multiple translational steps. Rotational angle ranged from 0° to 360° for all targets. For each trial, the operator was instructed to navigate the handle to a target position. A trial was considered successful when both the translational and rotational readings fall within a tolerance band of ± 2 mm and $\pm 5^\circ$ and remain within this band for two seconds. The mean time to target was reported as the primary performance indicator, computed separately for the two target groups. Additionally, the activity of both encoder channels were logged through the each trial to verify that simultaneous dual-axis input was employed during navigation.

5 Results and Discussion

Experiment 1 evaluated the repeatability of the translational encoder. The error across the five trials for the each of the targets is shown in Figure 7.

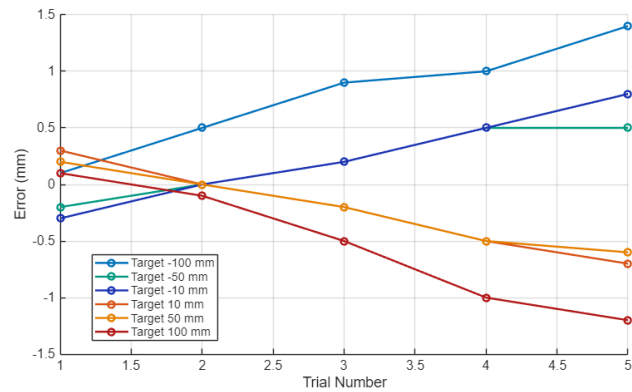


Figure 7: Translational measurement error plotted against trial number, displaying a trend where error increases with the trial number for both translational directions.

Across all target positions, a consistent small decrease in encoder-reported displacement accuracy was observed across successive trials, with a maximum recorded error of 1.4 mm for the 100 mm translational target. The standard deviation across trials ranged from 0.39 mm at the ± 10 mm positions to 0.56 mm at the ± 100 mm positions,

indicating a slight decrease in precision toward the stroke extremes.

These results suggest that the auto re-centring mechanism did not return the carriage to the exact same home position between trials. Instead, a systematic inaccuracy is introduced, where a small positional offset accumulates with each return cycle. This behaviour is likely attributed to mechanical effects in the belt-drive system. The static friction along the shaft-handle interface due to shaft surface roughness and at motor start-up resists initial motion, while minor belt slippage due to deformation or improper tensioning may further contribute to the positioning errors. As a result, the system fails to return the carriage precisely to the true home position but with a small offset.

Although the maximum observed error of 1.4 mm exceeds a tolerance band of ± 1 mm, the open loop control by the operator navigating to a target position compensates for the error. However, the problem for the systematic positional inconsistency with the auto re-centring system should still be addressed. A straightforward software mitigation would be to extend the return command by an empirically determined offset to compensate for the undershoot. A more robust long-term solution would be implement a control algorithm using the translational encoder to correct and adjust the handle to the same home position after each cycle.

Experiment 2 validated the multi-turn rotation capability of the master console handle.

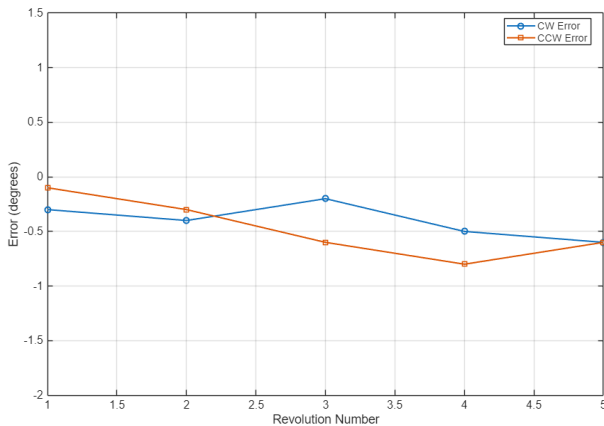


Figure 8: Angular measurement error plotted against revolution number. No significant trend or correlation is found between the angular error and number of revolutions, and error remains small within $\pm 1^\circ$.

Fig 8 shows the error for each successive revolution, where the error remains within 0.1° and 0.8° . Hence, it

can be concluded that all data points achieved the expected angular rotation value for both clockwise and counter clockwise trials within a $\pm 1^\circ$ tolerance. No discontinuities were observed at the 360° rollover boundary of the encoder, where the error remains consistent across all revolutions. The observed error therefore more likely attributable to operator precision and variability in confirming the reference mark position rather than limitations of the system itself. This suggests that the rotational encoding method employed in this paper provides a reliable and sufficiently accurate measurement of the angular rotation applied on the cylindrical handle.

Experiment 3 evaluated the navigational performance of the master console to arbitrary commanded positions. The time taken for each trial is presented in Fig 9, with the mean time calculated separately for the trials requiring one single stroke with translational target within 0 to 100 mm, and the trials requiring two steps where the translational target goes from 100 to 200 mm.

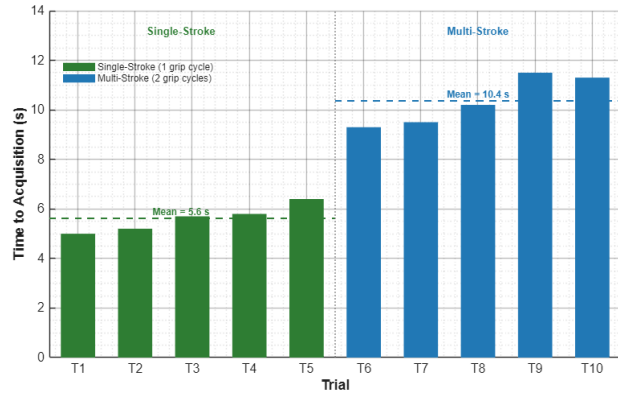


Figure 9: Time taken to reach target position plotted against trial number, with the green bars representing single-stroke trials and blue bars representing multi-stroke trials.

In Fig 9, the time taken for single-stroke trials ranges from 5 to 6.4s, with a mean of 5.6s. The time taken for trials requiring two strokes ranges from 9.3 to 11.3s, with a mean of 10.4s. The additional time between the two trial groups is attributable to the time taken for the auto re-centring process between steps, which is approximately 5s per additional cycle. These results suggest that the master console is capable of position navigation across its operational envelope, where all trials succeed in reaching the target position, including targets that exceed the maximum translational distance for a single stroke. The marginal increase in time for multi-stroke trials is expected from

the auto re-centring mechanism in comparison to joystick based controls and does not represent a functional limitation of the system. In the clinical context of RHC, this additional time is unlikely to be significant, as catheter advancement is performed in discrete steps rather than one continuous stroke, and the re-centring interval coincides with the natural pauses in manipulation. Additionally, simultaneous dual-axis input was confirmed across all trials, where both translational and rotational channels register concurrent displacement, validating the two separate degrees of freedom control of the mechanical design.

It should be noted that this experiment evaluates the motion input capability of the master console for the catheter replica control handle approach, and does not constitute a clinical feasibility study. The targets were defined as scalar positions with their translational distance and rotational angle required to reach the position. So, this experiment does not include the anatomical pathway constraints, as well as navigation of the catheter using feedback from real-time imaging. A more comprehensive validation of the master console's capability in a RHC setting requires a realistic virtual simulation environment or physical phantom testing. Such studies would allow for better evaluation of performance and feasibility for the RHC procedure, with analysis on trajectory consistency, procedure time across a complete RHC workflow, and operator usability feedback from experienced clinicians in catheter manipulation. These are identified as the primary direction for future experimental work.

6 Sustainability Analysis

This paper will discuss on the energy theme for sustainability considerations, especially in the context of the imaging modality choice of MRI-guided RHC relative to the conventional X-ray fluoroscopy-guided approach. The master console itself contributes negligibly to the energy consumption, where it is powered by a 24 W AC/DC adapter over a short RHC procedure time, amounting to approximately 0.02 kWh for a 1 hour procedure. Therefore, the master console and robotic catheter system's sustainability impact is dominated by the choice of imaging modality it enables, which is the shift to MRI guidance from fluoroscopy.

MRI scanners are one of the most energy-intensive de-

vices in a hospital, consuming an average of 19.9 kWh per examination, and a energy consumption around 100 MWh per scanner over the period of a year [28][29]. The total energy consumption increases when including the cooling energy consumption for its dedicated cooling systems, where the true energy cost per MRI examination increases to 35.8 kWh. In comparison, the energy consumption for different fluoroscopy scanners ranges from about 2280 to 14400 kWh over a year [30]. So, MRI scanners consume approximately 1 to 2 orders magnitude more energy annually than fluoroscopy systems, which is considerably higher. Similarly for emissions, the higher energy consumption of MRI contributes to a much greater carbon footprint compared to fluoroscopy.

Although MRI scanners have a much higher energy consumption, it helps reduce ionising radiation exposure from fluoroscopy operations, creating a energy trade-off problem. Radiation exposure poses health concerns such as increased risk of developing cancer, especially for interventional cardiologists who are consistently exposed to the radiation [31]. Replacing fluoroscopy with MRI guidance eliminates this risk for both the operator and patient. Additionally, MRI can provide superior soft tissue contrast for better imaging resolution. The benefits provided by MRI are real and important, therefore justifying its extra energy consumption.

Still, there is a path to improvement to reduce the energy consumption of MRI scanners. This is achieved through achieving better efficiency and power management. One example is the development in helium-free MRI magnet design, which operates at a much higher temperature and reducing the energy consumption required for the cooling system [32]. Also, another major improvement can be made by optimizing the operation scheduling and reducing idle time, as the scanner continues to consume a large amount of energy despite on stand-by mode. For the current prototype of the master console and teleoperated robotic system, the priority would be to ensure that the system enables more efficient procedures, leading to shorter procedure durations and thus lower MRI scanner energy consumption.

7 Conclusion

This paper presented the design and experimental validation of a cylindrical handle master console for a teleoperated MR-guided robotic RHC system. The relevance of this work lies in the development of better human-machine interfaces for teleoperated systems in the field of medical robotics. Master control console design that capitalize clinician experience remains an underexplored opportunity for improvement in efficiency and performance of the robotic system.

Adopting a catheter-replica approach, a single cylindrical handle and shaft interface is proposed to replicate the natural push-pull and rotation gestures of direct bedside catheter manipulation. The primary outcome of this work is a functional master console prototype with the following characteristics: infinite continuous rotation, 100 mm stroke for both push and pull directions with auto re-centring for infinite translational range, simultaneous dual-axis input, and a capacitive grip-sensing safety interlock, implemented within a low-cost assembly.

From experimental testing, the principal limitation of the current prototype is the auto re-centring mechanism, which introduces a systematic positional drift. This should be addressed through improvements in the mechanical design of the belt-drive system and control loop in the next design iteration. Also, the validation presented in this paper evaluates motion input capability, but clinical feasibility through realistic full procedure trials under anatomical pathway constraints and real-time imaging feedback remains for future work.

If validated in these settings, the cylindrical handle console approach has meaningful application potential for improving control intuitiveness. This translates to more precise catheter position control, reduced procedure time and lower cognitive load for clinicians through natural manipulation motions.

For further work, the next iteration of the prototype should address the aforementioned limitations. Then, it could be advanced at the system integration level, where the console can be connected to the slave robot for full workflow testing and evaluation. Finally, a formal ergonomic and human factor assessment involving clinician usability feedback is recommended to refine and validate the mechanical design and control mapping.

8 Declarations

8.1 Acknowledgements

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8.2 Use of AI

Claude Sonnet 4.6, developed by Anthropic, was used for assist with academic phrasing and proofreading throughout the writing process.

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